

SUSHMITHA ELLALA

Full Stack Developer

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PROFESSIONAL SUMMARY:

Experienced Full Stack Developer with practical knowledge of Django, Node.js, and Vue.js, creating responsive, scalable web apps. competent in handling SQL/NoSQL databases, working in Agile contexts, and integrating RESTful APIs. enthusiastic about improving development processes and developing user-focused digital solutions

EDUCATION:

Satavahana University, Karimnagar | Jun 2020 – May 2023
Bachelor of Computer Science

Skills

- | | |
|--|---|
| ✓ Languages: Python, JavaScript | ✓ Tools: Git, GitHub, Postman, VS Code |
| ✓ Frameworks: Node.js, Express.js, Django | ✓ Other: REST APIs, MVC Architecture |
| ✓ Frontend: HTML, CSS, Bootstrap, Vue.js | ✓ Agile, Source control |
| ✓ Databases: PostgreSQL, MySQL, MongoDB | ✓ Test driven development |

PROFESSIONAL EXPERIENCE:

Ibridge Digital

Full Stack Developer | July 2023 - September 2024

Internship : worked on creating a scalable multi-service e-commerce web platform that provides services including pet boarding, grooming, veterinary care, walking, and training to both people and businesses. Real-time communication, hierarchical staff-role administration, dynamic forms for medical modules, and subscription-based access are all supported by the system.

Infoware India

Node.js Developer | Oct 2024 - Present

PROJECTS

MediaWall – Digital Signage Management Platform

Node.js | Express.js | MongoDB | Socket.IO

- Designed and developed scalable RESTful backend APIs for a cloud-based multi-screen digital signage platform.
- Implemented real-time communication using Socket.IO for live content updates, alerts, and screen synchronization.
- Built JWT-based authentication and role-based access control (RBAC) to secure multi-user and multi-organization workflows.
- Designed and maintained a multi-tenant MongoDB database schema ensuring data isolation and horizontal scalability.
- Optimized backend performance by improving query execution, reducing response latency, and increasing system availability.
- Integrated third-party services and scheduling modules to support automated publishing and analytics reporting.

2. Coffee Shop Discovery & Management Platform

(Node.js | TypeScript | PostgreSQL)

Overview:

This project is a comprehensive backend ecosystem designed to connect coffee lovers with local cafes while empowering vendors with robust management tools. It serves as a multi-tenant platform where users can discover new shops, browse complex menus with deep customization options, and handle secure transactions. For business owners, it acts as a mini-ERP, handling everything from inventory and operating hours to detailed sales analytics. The system allows for rapid onboarding of hundreds of businesses through automated scraping and intelligent bulk import systems.

Key Features:

- **Intelligent Bulk Import:** An Excel processing system that validates thousands of rows, auto-generates credentials, and maps raw GPS coordinates to structured addresses (State, City, Area) using reverse geocoding.
- **Complex E-commerce Logic:**
- **Dynamic Menu Architecture:** Supports deeply nested menu items (e.g., Categories → Sub-Categories → Variants → Customization Groups → Modifiers), allowing for unlimited product combinations.
- **Split Payments & Invoicing:** Handles complex transaction flows where payments may need to be split between the platform and vendors, with automated invoice generation for every order.
- **Shop Management & Logistics:**
- **Flexible Scheduling:** distinct logic for managing split-shift operating hours (e.g., 8 AM–12 PM & 4 PM–9 PM) and holiday exceptions.
- **Role-Based Dashboards:** Granular access control for Super Admins, Shop Managers, and Customers, providing tailored analytics (Average Order Value, Sales Trends) for each role.
- **Location Intelligence:** Hierarchical management of location data (Country → State → District → City → Area) to support precise search filtering and logistics.

My Roles & Responsibilities

- **Backend Development & API Architecture:**
 - Designed and maintained scalable RESTful APIs using Node.js, TypeScript, and Express, ensuring type safety and code modularity across the application.
 - Managed the PostgreSQL database schema using Sequelize ORM, optimizing complex queries for menu retrieval and order history.
- **Data Engineering & Automation:**
 - **Developed the Google Maps Scraper:** Engineered a robust scraping script to extract data for 150+ shops alongside their menus and media, handling anti-bot measures and dynamic DOM elements.
 - **Built the Bulk Import System:** Created the logic to parse Excel files, implementing strict data validation (checking for duplicates, invalid formats) and auto-creating missing location hierarchies (e.g., creating a new "City" record if it didn't exist) during upload.
- **Third-Party Integrations & Debugging:**
 - **Payment Gateway Integration:** Resolved critical issues in the live production environment for payment callbacks (webhooks) and implemented error handling for failed split-payment scenarios to ensure data consistency.
 - **Media Management:** Integrated AWS S3 for secure storage of shop logos and menu images, debugging file upload streams and associating media with distinct database entities.
- **Feature Optimization:**
 - Refactored the Shop Dashboard APIs to align data calculations with domain-level analytics, ensuring business owners see accurate real-time metrics.
 - Implemented logic for complex shop updates, such as handling CRUD operations for multiple "Shop Hours" entries (soft deletes/upserts) within a single transaction.

3. Enterprise Healthcare Management System (HIMS)

(Node.js | Express.js | MySQL |)

Overview:

Developed a comprehensive, scalable Hospital Information Management System (HIMS) designed to digitize and streamline hospital operations. The system features a modular architecture supporting multiple user roles (Doctors, Patients, Nurses, Lab Technicians, Administrators) with specialized workflows for Patient Management, Laboratory Information Systems (LIS), Pharmacy Inventory, and Asset Tracking.

Key Features & Modules

1. Core Architecture & Security

- **Role-Based Access Control (RBAC):** Implemented distinct, secure distinct portals for Super Admins, Doctors, Nurses, Patients, and Technicians using custom middleware (`adminVerifyToken`, `verifyToken`).
- **Multilingual Support:** Built a dynamic localization engine allowing medical records and prescriptions to be stored and retrieved in multiple languages (English, French, etc.) using JSON-based database fields.

2. Doctor & Patient Portals

- **Doctor Dashboard:** Real-time analytics, appointment management, and direct access to patient medical history.
- **Electronic Medical Records (EMR):** Digital management of prescriptions, medical history, and lab test orders.
- **Patient Interaction:** Patients can view their medical history, download prescriptions, and access lab reports securely.

3. Laboratory Information System (LIS)

A fully-featured module for managing pathology and diagnostic labs:

- **Sample Management:** Tracking sample collection, processing, and status updates.
- **Test Management:** Configurable test services, parameters, and result entry.
- **Inventory & Equipment:** Tracking lab inventory (GRN, Suppliers) and equipment maintenance schedules.
- **Reporting:** Automated generation of patient lab reports and unified test results.

4. Pharmacy & Asset Management

- **Pharmacy Inventory:** Real-time tracking of medicine stock, expire dates, and unit-of-measure (UOM) management.
- **Asset Tracking:** Lifecycle management of hospital assets, including purchase orders, vendor management, and maintenance logs.

5. Advanced Integrations

- **WhatsApp Notifications:** Integrated a custom notification channel using Puppeteer to send automated updates (appointments, reports) to patients via WhatsApp.
- **Cloud Storage:** Seamless integration with AWS S3 for uploading and retrieving patient documents, X-rays, and profile images.